

How Probability Is Used to Detect Asteroid Surface-Colour Variation

Equations, statistical workflow, simulations, and results

Based on Humes and Agarwal (2026)

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1. Purpose of the probability method

The scientific question is:

Do different regions of an unresolved asteroid reflect different relative amounts of light in different wavelength bands?

The telescope does not produce an image of the asteroid's surface.

It measures one total brightness value through a filter at each observation time.

The measured brightness changes for many ordinary reasons:

- the asteroid rotates;
- its projected area changes;
- its distance from the Sun changes;
- its distance from Earth changes;
- the solar phase angle changes;
- different latitudes and longitudes become illuminated and visible;
- measurements contain random and systematic errors.

Therefore, an unusual measurement alone is not evidence of a coloured surface region.

The probability method asks a more specific question:

If the asteroid were uniformly coloured, how likely would it be to obtain residual patterns this different between two filters purely because of sampling and noise?

The method converts this question into a statistical significance score.

2. The hypotheses being tested

2.1 Null hypothesis

The null hypothesis is:

$$H_0 : \text{the asteroid is spatially uniform in colour.}$$

The asteroid may still have an overall mean colour.

For example, its average $g - r$ value may be nonzero.

The assumption only says that one longitude is not intrinsically redder or bluer than another.

2.2 Alternative hypothesis

The alternative hypothesis is:

$$H_1 : \text{at least one surface region has different wavelength-dependent reflectance.}$$

The test does not directly photograph that region.

It searches for a repeatable, band-dependent brightness departure connected to body-fixed viewing geometry.

3. Data and models used by the test

For each observation, the analysis needs:

- epoch t ;
- measured magnitude $m_{B,\text{obs}}(t)$;
- wavelength band B ;
- magnitude uncertainty;
- asteroid-to-Sun vector;
- asteroid-to-observer vector;
- heliocentric and observer distances;
- asteroid shape model;
- rotation period;
- spin-axis orientation;
- initial rotational phase and reference epoch;
- band-dependent phase-function model.

For real work, survey photometry may come from ZTF.

Ephemerides may be calculated using kete or another ephemeris service.

A selected real asteroid's shape and spin may come from DAMIT.

The model predicts what the asteroid should look like if its surface colour is uniform.

4. Step 1 - Predict the normal brightness

For every band B and time t , the model predicts:

$$m_B(t).$$

The prediction accounts for ordinary effects such as:

- distance;
- phase angle;
- asteroid shape;
- rotational orientation;
- viewing direction;
- illumination direction;
- mean band-dependent reflectance.

The physical shape and spin must be the same in every wavelength band.

However, absolute magnitude and phase-function parameters may differ by band.

That allows the model to represent the asteroid's average colour without allowing local colour patches.

5. Step 2 - Calculate residual magnitude

For each observation, calculate the difference between measured and predicted magnitude:

$$\Delta m_B(t) = m_{B,\text{obs}}(t) - m_B(t).$$

$\Delta m_B(t)$ is the residual.

5.1 Interpretation of residual sign

Because the magnitude scale runs backwards:

- $\Delta m_B = 0$ means observation and model agree;
- $\Delta m_B > 0$ means the object was fainter than predicted;
- $\Delta m_B < 0$ means the object was brighter than predicted.

5.2 Numerical example

Suppose an R -band observation is:

$$m_{R,\text{obs}} = 14.20$$

and the uniform model predicts:

$$m_R = 14.32.$$

Then:

$$\Delta m_R = 14.20 - 14.32 = -0.12 \text{ mag.}$$

The asteroid is 0.12 magnitude brighter than predicted in that observation.

The corresponding flux ratio is:

$$\frac{F_{\text{obs}}}{F_{\text{model}}} = 10^{-0.4\Delta m_R}.$$

For $\Delta m_R = -0.12$:

$$\frac{F_{\text{obs}}}{F_{\text{model}}} = 10^{0.048} \approx 1.117.$$

The observed flux is therefore approximately 11.7% greater than the model prediction.

5.3 Why residuals are useful

Raw magnitudes are dominated by normal rotational and geometric variation.

Residuals attempt to isolate what the uniform model did not explain:

$$\text{residual} \approx \text{surface signal} + \text{measurement error} + \text{model error}.$$

A residual is not automatically a spot detection.

The unexplained signal must also repeat with surface orientation and behave differently by wavelength.

6. Step 3 - Describe which part of the asteroid contributes

Let:

$$\hat{\mathbf{E}}_0(t)$$

be the unit vector from the asteroid towards the Sun, expressed in the asteroid's rotating frame.

Let:

$$\hat{\mathbf{E}}(t)$$

be the unit vector from the asteroid towards the observer.

The Sun direction tells us which side is illuminated.

The observer direction tells us which side is visible.

The authors combine them using:

$$\tilde{\mathbf{E}}(t) = \frac{1}{2} [\hat{\mathbf{E}}_0(t) + \hat{\mathbf{E}}(t)].$$

This is the average illumination-viewing direction.

It approximately identifies the surface direction favoured by both illumination and visibility.

6.1 Why not use the Sun vector alone?

A strongly illuminated region contributes nothing if it faces away from Earth.

6.2 Why not use the observer vector alone?

A visible region contributes no reflected sunlight if it is on the night side.

6.3 Components of the geometry vector

In the asteroid-fixed coordinate frame:

$$\bar{\mathbf{E}} = (\bar{E}_x, \bar{E}_y, \bar{E}_z).$$

These three numbers describe the geometry at one epoch.

The vector changes with time because the asteroid rotates and its orbital geometry changes.

7. Step 4 - Find the direction most associated with residuals

A surface feature can lie in an arbitrary direction.

It need not align with the model's x , y , or z axis.

The authors therefore use multiple linear regression:

$$\Delta m = k_x \bar{E}_x + k_y \bar{E}_y + k_z \bar{E}_z + \epsilon.$$

Here:

- k_x , k_y , and k_z are fitted weights;
- ϵ is residual variation not explained by the linear model.

In matrix notation:

$$\mathbf{y} = \mathbf{X}\mathbf{k} + \epsilon.$$

The least-squares estimate is conceptually:

$$\hat{\mathbf{k}}_{\text{LS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y},$$

when the design matrix has suitable rank.

The coefficient vector is normalized:

$$\hat{\mathbf{k}} = \frac{(k_x, k_y, k_z)}{\sqrt{k_x^2 + k_y^2 + k_z^2}}.$$

$\hat{\mathbf{k}}$ is the direction of maximum residual variation.

It indicates the body-fixed direction where the uniform model performs worst.

This is not a detailed surface map.

It is a statistical direction that can guide follow-up observations.

8. Step 5 - Reduce the 3D geometry to one predictor

For each epoch, project the average geometry vector onto the fitted direction:

$$\mathcal{L}(\bar{\mathbf{E}}) = \bar{\mathbf{E}} \cdot \hat{\mathbf{k}}.$$

Expanded:

$$\mathcal{L} = \bar{E}_x \hat{k}_x + \bar{E}_y \hat{k}_y + \bar{E}_z \hat{k}_z.$$

This changes three geometry coordinates into one scalar predictor.

Interpretation:

- large positive $\mathcal{L}$: geometry points broadly towards $\hat{\mathbf{k}}$;
- large negative $\mathcal{L}$: geometry points broadly opposite $\hat{\mathbf{k}}$;
- $\mathcal{L} \approx 0$: geometry has little projection along that direction.

The test can now ask whether magnitude residual changes monotonically with $\mathcal{L}$.

9. Step 6 - Calculate Spearman correlation separately in each band

For every band B , the method calculates:

$$\rho_B = \text{corr}_{\text{Spearman}}[\mathcal{L}, \Delta m_B].$$

Spearman correlation measures monotonic association using ranks.

It ranges from -1 to $+1$.

9.1 Interpretation

- $\rho_B \approx +1$: residual tends to increase as $\mathcal{L}$ increases;
- $\rho_B \approx -1$: residual tends to decrease as $\mathcal{L}$ increases;
- $\rho_B \approx 0$: no clear monotonic relationship.

9.2 Rank-difference formula

When there are no tied ranks, Spearman correlation can be calculated as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)},$$

where:

- n is the number of paired observations;
- d_i is the difference between the rank of $\mathcal{L}_i$ and the rank of Δm_i .

With ties, software normally assigns average ranks and computes Pearson correlation between the rank variables.

9.3 Simple example

Suppose the ranks of $\mathcal{L}$ are:

$$1, 2, 3, 4, 5.$$

Suppose the residual ranks are:

$$1, 2, 4, 3, 5.$$

The differences are:

$$d = (0, 0, -1, 1, 0).$$

Therefore:

$$\sum d_i^2 = 2.$$

Then:

$$\rho = 1 - \frac{6(2)}{5(5^2 - 1)} = 1 - \frac{12}{120} = 0.90.$$

The two variables have a strong positive monotonic relationship.

9.4 Why use Spearman correlation?

The brightness contribution of a partly visible spot need not be exactly linear.

Spearman correlation can identify a monotonic pattern without requiring an exact straight-line relationship.

10. Step 7 - Compare the two wavelength bands

Suppose the uniform model has a geometry-related error caused by an imperfect shape.

That physical shape error may affect both bands similarly.

For example:

$$\rho_V = -0.62, \quad \rho_R = -0.65.$$

The correlations are strong but nearly equal.

This does not provide strong evidence for colour variation.

Now suppose:

$$\rho_V = -0.08, \quad \rho_R = -0.72.$$

The geometry relationship is much stronger in R than in V .

That is more consistent with a surface feature having wavelength-dependent reflectance.

The method must determine whether this difference is larger than expected from correlation-estimation uncertainty.

11. Step 8 - Fisher transformation of each correlation

Correlation coefficients are limited to the interval $[-1, 1]$.

Their sampling distribution is not normally distributed, especially near the boundaries.

The Fisher transformation is:

$$z_B = \frac{1}{2} \ln \left(\frac{1 + \rho_B}{1 - \rho_B} \right). \quad (2)$$

Equivalently:

$$z_B = \operatorname{atanh}(\rho_B).$$

The transformed correlation is more nearly normally distributed for adequate sample sizes. Its approximate variance is:

$$\operatorname{Var}(z_B) \approx \frac{1}{n_B - 3}.$$

Its approximate standard error is:

$$\operatorname{SE}(z_B) \approx \frac{1}{\sqrt{n_B - 3}}.$$

11.1 Examples

For $\rho = 0$:

$$z = 0.$$

For $\rho = 0.5$:

$$z = \frac{1}{2} \ln(3) \approx 0.5493.$$

For $\rho = -0.5$:

$$z \approx -0.5493.$$

12. Step 9 - Significance of the difference between bands

For independent observations in bands B_1 and B_2 , the paper uses:

$$z_{B_1, B_2} = \frac{z_{B_1} - z_{B_2}}{\sqrt{\frac{1}{n_{B_1} - 3} + \frac{1}{n_{B_2} - 3}}}. \quad (1)$$

Here:

- z_{B_1} and z_{B_2} are Fisher-transformed correlations;
- n_{B_1} and n_{B_2} are observation counts in the two bands;
- the denominator is the approximate standard error of the difference.

12.1 Why the two variance terms are added

For independent estimates:

$$\text{Var}(z_1 - z_2) = \text{Var}(z_1) + \text{Var}(z_2).$$

Using the Fisher approximation:

$$\text{Var}(z_1 - z_2) \approx \frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}.$$

The square root converts variance into standard error.

12.2 Meaning of the final score

The numerator measures the observed difference between band correlations.

The denominator measures how much that difference should fluctuate because the samples are finite.

Therefore:

$$\text{significance} = \frac{\text{observed difference}}{\text{expected statistical uncertainty}}.$$

A larger absolute score means the band correlations are more difficult to explain as equal.

The paper selects potential detections using:

$$z_{B_1, B_2} > 5.$$

This is the paper's 5σ detection threshold.

13. Complete worked significance example

Assume:

$$\rho_V = -0.10, \quad \rho_R = -0.70,$$

with:

$$n_V = n_R = 100.$$

13.1 Transform the V -band correlation

$$z_V = \frac{1}{2} \ln \left(\frac{1 - 0.10}{1 + 0.10} \right)$$

$$z_V = \frac{1}{2} \ln(0.81818) \approx -0.1003.$$

13.2 Transform the R -band correlation

$$z_R = \frac{1}{2} \ln \left(\frac{1 - 0.70}{1 + 0.70} \right)$$

$$z_R = \frac{1}{2} \ln(0.17647) \approx -0.8673.$$

13.3 Calculate the difference

$$z_V - z_R = -0.1003 - (-0.8673) = 0.7670.$$

13.4 Calculate the standard error

$$SE = \sqrt{\frac{1}{97} + \frac{1}{97}}$$

$$SE \approx 0.1436.$$

13.5 Calculate the final statistic

$$z_{V,R} = \frac{0.7670}{0.1436} \approx 5.34.$$

Because:

$$5.34 > 5,$$

this example passes the paper's threshold.

It becomes a candidate detection of wavelength-dependent surface heterogeneity.

It is not yet proof.

Model errors, survey systematics and multiple testing must still be investigated.

14. Why probability is tested using simulations

The analytical z formula alone does not describe every complication in the full algorithm.

The regression direction is estimated from data.

Sampling geometry is irregular.

The spot can have different size, location and contrast.

Model parameters can be wrong.

The authors therefore use Monte Carlo simulations.

Monte Carlo analysis means repeating the experiment many times with randomly selected asteroid properties and errors.

15. Synthetic datasets used for probability calibration

For each base synthetic asteroid, the authors produce two datasets.

15.1 Spotted dataset

One wavelength band contains a region with enhanced reflectance.

This dataset should be detected.

15.2 Uniform dataset

The surface is uniformly coloured.

This dataset should not be detected.

For 1,000 base asteroids:

$$N_{\text{datasets}} = 1000 \text{ spotted} + 1000 \text{ uniform} = 2000.$$

The same test is applied to both groups.

16. Probability distributions used in the simulation

16.1 Uniform distributions

A uniform distribution gives equal density across a selected interval.

The notation is:

$$X \sim \mathcal{U}(a, b).$$

Examples in the simulation include:

- rotation period sampled between 0.1 and 2 days;
- initial rotation angle sampled from 0 to 360°;
- spot angular size sampled from 20° to 90°;
- spot reflectance enhancement sampled from 1.2 to 2.0;
- G_{12}^* phase parameter sampled between 0 and 1.

Pole and spot-centre directions are sampled uniformly over a sphere.

Correct spherical sampling requires uniform longitude and uniform $\sin(\text{latitude})$, not uniform latitude.

16.2 Gaussian measurement noise

Synthetic magnitude noise is represented as:

$$\epsilon_m \sim \mathcal{N}(0, \sigma_m^2).$$

The noisy magnitude is:

$$m_{\text{noisy}} = m_{\text{true}} + \epsilon_m.$$

The mean is zero, so noise does not intentionally brighten or dim the entire dataset.

σ_m controls its size.

16.3 Gaussian model-parameter perturbation

For a true parameter q_{true} :

$$q_{\text{assumed}} \sim \mathcal{N}(q_{\text{true}}, \sigma_q^2).$$

The standard deviation is gradually increased to determine how inaccurate the parameter can become before detection performance fails.

The pages shown evaluate measurement noise and begin the period-error analysis.

17. True-positive and false-positive rates

For spotted asteroids, the detection or true-positive rate is:

$$\text{TPR} = \frac{N(\text{spotted datasets with } z > 5)}{N(\text{spotted datasets})}.$$

For uniform asteroids, the false-positive rate is:

$$\text{FPR} = \frac{N(\text{uniform datasets with } z > 5)}{N(\text{uniform datasets})}.$$

17.1 Example

Suppose 920 of 1,000 spotted datasets pass:

$$\text{TPR} = \frac{920}{1000} = 0.92.$$

Suppose 3 of 1,000 uniform datasets pass:

$$\text{FPR} = \frac{3}{1000} = 0.003.$$

The method detects 92% of spots under those conditions and incorrectly flags 0.3% of uniform asteroids.

17.2 What detection rate does not mean

A 92% detection rate does not mean that one individual detection has a 92% probability of being real.

It is a population-level performance measure under the simulated conditions.

An individual posterior probability would additionally require prior probabilities and explicit likelihood modelling.

18. Probability workflow shown in Figure 1

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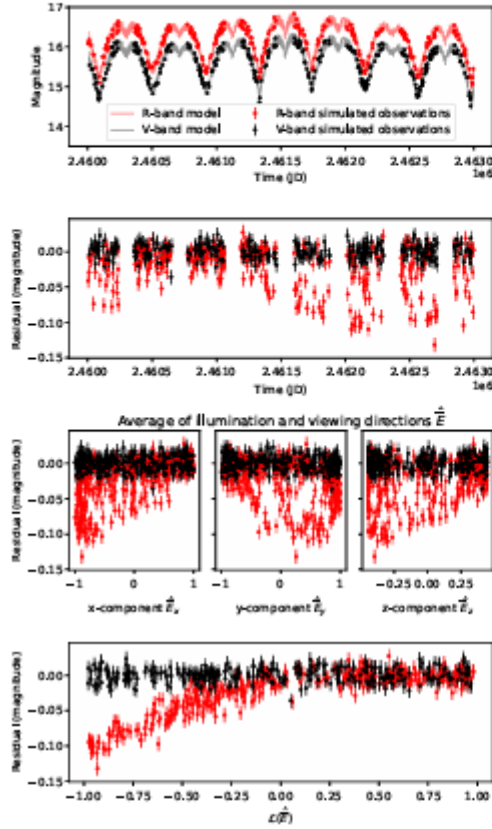


Fig. 1. Graphical summary of the steps of the statistical test for surface color variation for a simulated asteroid (see Section 3 and Appendix A for a description of the photometric models) in the V and R bands. The simulated asteroid contains a large high albedo spot visible in the R band. First, a uniformly colored photometric model was fit to the simulated photometry. Second, the residuals (Δm) in predicted magnitude between the photometric model and the simulated photometry were computed. At this point, there were obvious discrepancies between the performance of the model in predicting the V-band and R-band, but it was unclear whether these discrepancies are associated with the viewing geometry of the asteroid. Third, the residuals in magnitude were plotted as a function of the x, y, z -components of the subsolar direction in the rotating reference frame of the asteroid. Fourth, multiple linear regression was used to identify the linear transformation $\mathcal{L}$ of $\hat{E}_x, \hat{E}_y, \hat{E}_z$ to a single predictor variable $\mathcal{L}(\hat{E})$ that maximizes the correlation between $\mathcal{L}(\hat{E})$ and Δm . The final step of the test (not pictured) was to compute the significance of the difference in the correlation coefficient between $\mathcal{L}(\hat{E})$ and Δm for the two bands using the Fisher z -transformation.

asteroid. We then computed the result of the test for surface heterogeneity assuming a “perturbed” photometric model, which differs from the truth model used as input to generate the synthetic photometry in a single parameter, the value of which is drawn from a (typically Gaussian) probability distribution with a mean equal to the true value of the parameter and standard deviation σ . By observing how the spot detection (correct detection of a spot in the photometry of a “spotted” asteroid) and false

positive rates (rate of incorrectly detecting a spot in the photometry of uniformly colored models) across all 2000 synthetic datasets change as the value of σ increased, we quantified the required accuracy for each parameter in the photometric model using a Monte Carlo approach. For each synthetic asteroid, a total of 73 perturbed models, differing from the truth model in a single parameter, were generated.

Our general approach to the photometric modeling of synthetic asteroids closely followed the general technique used to compute the direct problem of modeling the light curves of convex objects employed to solve the convex inversion problem (Kasalainen et al. 2001a; Kasalainen & Torppa 2001; Āurech et al. 2010). A similar approach to producing synthetic light curves from a discretized triangular mesh is described in Lu et al. (2014). We included minor modifications to these methods in our model to account for multiband observations and spatial variations in photometric properties, as detailed in the appendix. To summarize, the following inputs to the model were required to obtain synthetic photometry at a set of times t in wavelength bands B :

- For each time of interest t , the light time corrected asteroid-observer and asteroid-Sun vectors in ecliptic coordinates (to compute the asteroid’s reduced magnitude, E , and E_0).
- The rotational state of the asteroid (including the ecliptic coordinates of the rotational pole, rotational period, initial rotation angle, and reference epoch) to convert the asteroid-observer (E) and asteroid-Sun vectors (E_0) from ecliptic coordinates to the rotational frame of the asteroid.
- A convex triangular mesh representing the shape model of the asteroid.
- A disk function (scattering law) $S(\mu_0(t), \mu(t))$ describing the dependence of flux on the incidence and exitance angles μ_0 and μ .
- For each wavelength band of interest B ,
 - An absolute magnitude $H_{0,B}$ representing the observed magnitude under idealized conditions with both the sub-solar and sub-observer points located at $(0^\circ, 0^\circ)$ longitude and latitude,
 - The wavelength-dependent phase function $f_B(\alpha)$,
 - A reflectance map assigning a reflectance factor R_T to each facet of the shape model.

In generating ephemerides for the synthetic asteroids, we selected 1000 asteroids each at random from the populations of MBAs, NEAs, and TNOs with $H_{max} < 15$ as retrieved using the JPL Horizons database query. We aimed to evenly sample the actual distributions of orbital elements of these populations. We used the *kef* (Dahlen et al. 2025) package to propagate the asteroids’ orbits and compute the heliocentric and geocentric distances r and δ , as well as E and E_0 for an Earth-based observer for a total of 300 randomly sampled dates per wavelength band over the course of 3000 days. To simulate the solar elongation restraints of a ground-based telescope, we selected dates only among those intervals where the elongation of the asteroid as viewed from Earth exceeded 45° . Rotational pole orientations were chosen from the uniform distribution on a sphere. Rotational periods from a uniform distribution ranging from 0.1 to 2 days represented the majority of asteroids rotating slower than the spin barrier (Warner et al. 2009) while excluding ultra-slow rotators, which are likelier to be tumblers in more complex rotational states (Zhou et al. 2025). The initial rotation angle relative to the coordinate system of the asteroid was randomized between $0 - 360^\circ$ at the reference epoch. We modeled the shape of each asteroid as a triangulated mesh approximating an ellipsoid with asymmetric semiaxes,

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Figure 1 demonstrates one spotted synthetic asteroid.

Panel 1: simulated magnitude and fitted models

The oscillating curves mainly arise from shape and rotation.

The uniform models attempt to predict those ordinary changes.

Panel 2: residual versus time

The affected band's residuals show stronger departures from zero.

Panel 3: residual versus geometry components

Residuals are compared with the x , y , and z components of the average illumination-viewing vector.

Panel 4: residual versus $\mathcal{L}$

The regression-selected projection organizes the affected-band residuals into a clear monotonic trend.

Final probability step

The final Fisher comparison is not plotted in Figure 1.

It calculates whether the difference between the two band correlations is statistically significant.

19. Results by asteroid population and spot size

The paper reports the following detection rates:

Spot-size interval	NEAs	MBAs	TNOs
$80^\circ < \phi_{\text{spot}} \leq 90^\circ$	1.00	1.00	0.99
$70^\circ < \phi_{\text{spot}} \leq 80^\circ$	1.00	1.00	0.98
$60^\circ < \phi_{\text{spot}} \leq 70^\circ$	1.00	1.00	0.96
$50^\circ < \phi_{\text{spot}} \leq 60^\circ$	1.00	1.00	0.93
$40^\circ < \phi_{\text{spot}} \leq 50^\circ$	0.99	1.00	0.92
$30^\circ < \phi_{\text{spot}} \leq 40^\circ$	0.99	0.99	0.85
$20^\circ < \phi_{\text{spot}} \leq 30^\circ$	0.91	0.96	0.74

Interpretation

Large spots affect a greater fraction of disk-integrated brightness and are detected more often.

TNO detection is lower, particularly for small spots, because the 3,000-day survey samples a more limited fraction of their very long orbital and seasonal geometry.

The authors subsequently emphasize MBAs when isolating parameter sensitivities.

20. Result 1 - Number of observations

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detections of color changes associated with seasonal variations in aspect angle are possible only for objects which can be observed over many seasons, the total duration of the survey limits detections of latitudinal variation in color to those objects with orbital timescales less than or equal to the total duration of the survey. This is the primary factor strongly limiting the detection rate for TNOs: the test is simply not able to detect color variations on regions of the object not visible and illuminated during the duration of the survey, and this explains why smaller spots are more likely to be missed. As a result, detections of latitudinal variation in color are limited to asteroids in the Main Belt or closer. The situation is slightly more complicated for NEAs. Due to elongation constraints, observing these asteroids at all is only possible during certain portions of their orbital period, inherently limiting the possible aspect angles at which they are able to be observed, resulting in their intermediate detection rates. Additional information (such as brightness constraints and observation cadence) would be required to assess the dependence of detection rates on orbital elements in greater detail for a particular survey strategy. Given the expected timescales for present and near-future photometric surveys are on the order of the orbital timescales of MBAs, in subsequent tests, we limited our analysis to the MBA subpopulation only in order to derive upper bounds on the detection rate as a function of each perturbed variable. However, the test for heterogeneity is still applicable to TNO and NEA populations so long as the observations in each band samples the same distribution of latitudes and longitudes, to avoid false positive detections (see discussion in next section). In this case, the test is only sensitive to heterogeneities within the common range of latitudes and longitudes sampled by both bands.

4.2. Minimum number of observations

We evaluated the effect of varying the total number of observations of a given asteroid on the detection rate of surface spots. First, we assumed each band was observed an equal number of times. We computed z_{B_1, B_2} for a subset of the synthetic observations. Using a detection threshold of 5σ , we computed the true detection and false positive rate for the spotted and uniform synthetic asteroids respectively as a function of the number of times the asteroid was observed.

The results are shown in Figure 3. For all spot sizes and reflectance enhancement factors, the detection rate increases as more observations are taken. Spots with smaller angular sizes are more difficult to detect with a small number of observations, as the likelihood of failing to observe the asteroid when a small spot is illuminated and visible is relatively high when a limited number of observations are available. The false positive rate is $<1\%$ and decreases as more observations are taken. Based on the results, we estimate a conservative minimum of 100–150 observations total are needed to detect 90% of spots (depending mainly on spot size).

While we assumed the asteroid would be observed an equal number of times per bandpass, this is likely to not be the case for real data. Next, we evaluated the effect of varying the number of observations in each band independently. The results are shown in Figure 4. In general, the detection rate increases as the number of observations in either band increases. However, there is a slight asymmetry: increasing the number of observations in Band 2 increases the detection rate more than increasing the number of observations in Band 1 does. This is because the spot is only present in Band 2: the reflectance in Band 2 is not uniform as it is in Band 1, so the uniform albedo model performs

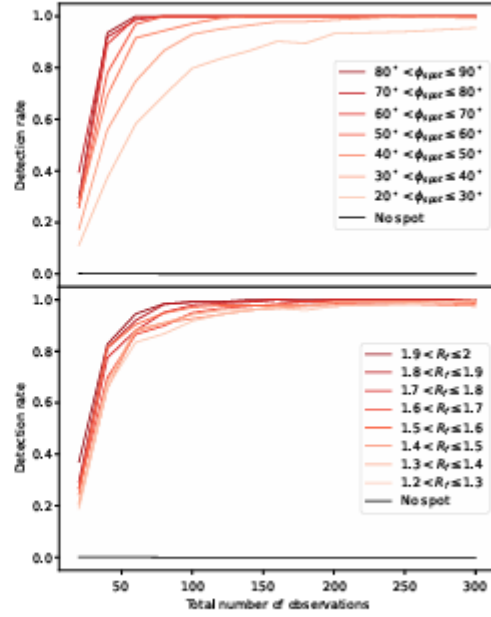


Fig. 3. Detection and false positive rates as a function of the number of times an asteroid is observed. Detection rates depend on spot properties, including the angular size of the spot ϕ_{spot} and the reflectance enhancement factor within the spot R_f .

more poorly for Band 2. The discrepancy between the model and observations is more readily apparent as more Band 2 observations are included. With real observations, we do not know a priori which band an albedo spot is present in, if at all, so the band that the uniform model performs best for will be dominated by the band with the most observations and/or the smallest error bars. Therefore the detection rates for the case where the number of observations in Band 1 exceeds the number of observations in Band 2 provide a more realistic estimate of the detection rate for situations in which an asteroid is not observed an equal number of times in each bandpass.

The performance of the test with low numbers of observations, especially the elevated false positive rate when only one band is well sampled (>30 observations), illustrates an additional criterion for the test's application. In addition to requiring a minimum number of observations, the distribution of subsolar and sub-observer latitudes and longitudes sampled by each band should be similar. Potential false positives can arise if a region of the asteroid is particularly poorly modeled by the assumed photometric model and this region is only (or predominantly) sampled in only one band. In this case, without sufficient data for the problematic region in the second band, it is impossible to distinguish between poor model performance due errors in assumed parameters (e.g., an inaccurate shape model) versus true surface color heterogeneity. When a single band only has a few observations, it is more likely that those observations do not cover the same range of subsolar and sub-observer latitudes and longitudes sampled by the more well observed band. While sampling the full range of possible subsolar and sub-observer latitudes and longitudes improves overall detection rates, for

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Figure 3 plots detection rate against total observation count when the two bands contain equal numbers of measurements.

The main findings are:

- detection improves as observations increase;
- large spots need fewer observations;
- strong-reflectance spots need fewer observations;

- approximately 100 to 150 total observations produce nearly complete detection for the simulated spot range under ideal conditions;
- this corresponds approximately to 50 to 75 observations per band when counts are balanced;
- the no-spot false-positive curve stays close to zero.

The result is not a universal guarantee for real ZTF data.

It assumes accurate model parameters and simulated observing conditions.

21. Result 2 - Imbalanced filter observations

Humes, O., and Agarwal, J.: A&A, 711, A209 (2026)

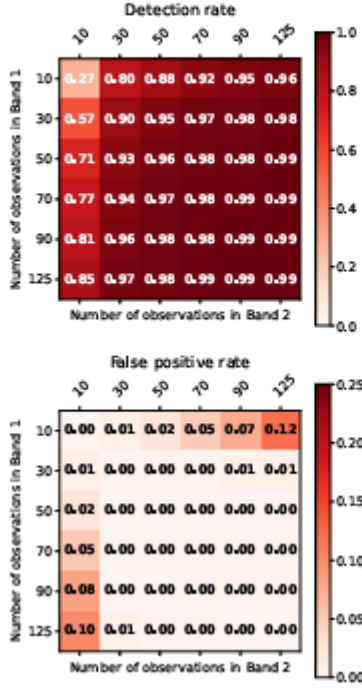


Fig. 4. Detection and false positive rates as a function of the number of observations in each of the two bands. The synthetic asteroid has a uniform albedo in Band 1 but displays an albedo spot in Band 2.

some objects, particularly those beyond the Main Belt, obtaining such observations would require observations spanning infeasibly long timescales. However, the test can still be applied by restricting the test to those photometric data that sample a common distribution of latitudes and longitudes.

4.3. Sensitivity to random measurement error

We evaluated the effect of adding random Gaussian noise to the simulated magnitudes to approximate the effect of measurement errors arising from real-world observing conditions. We computed z_{B_1, B_2} for observations with increasing levels of noise in measured magnitudes, assuming all other model parameters had been accurately inferred. Using a detection threshold of 5σ , we computed the true detection and false positive rate for the spotted and uniform synthetic asteroids respectively as a function of the standard deviation of the added Gaussian noise.

The results are shown in Figure 5. For all spot sizes and reflectance enhancement factors, the true detection rate decreases to near 0 as the standard deviation of the added noise increases. The false positive rate remains near 0 for all noise levels. Adding random Gaussian noise to the measured magnitude increases scatter among the model residuals in the Δm_B direction, without any dependence on $\hat{E}$. The added scatter effectively reduces the measured correlation coefficient between $\hat{E}$ and Δm_B in bands where a correlation is present without introducing spurious dependencies on $\hat{E}$, resulting in the observed dependence of the detection rate and the independence of false positive rate

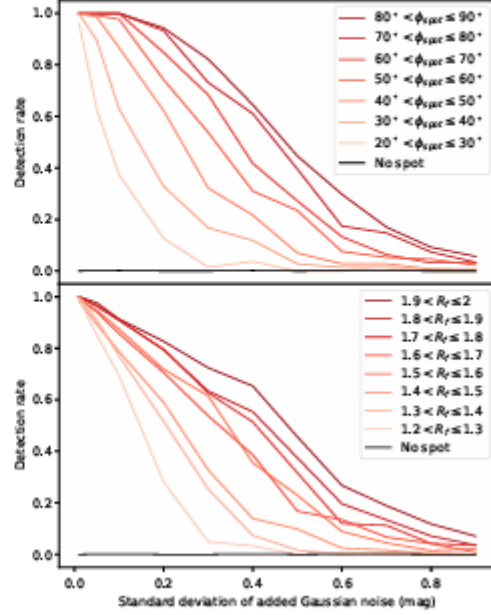


Fig. 5. Detection and false positive rates as a function of increased added Gaussian noise. Detection rates depend on spot properties, including the angular size of the spot (ϕ_{spot}) and the reflectance enhancement factor within the spot (R_f).

on the level of increased noise. Spots that are larger in size or with stronger flux enhancements are more readily detectable at high noise levels. The detection rate of spots with widths less than $20-30^\circ$ drops dramatically as a function of noise, reaching $<50\%$ for observations with >0.1 mag uncertainties. The $\phi_{spot} > 20^\circ$ limit gives a realistic lower bound on the minimum angular size of regional color variations that could be inferred from whole-disk photometric measurements. Similarly, the detection rate of spots with reflectance enhancement factors of $R_f < 1.2$ is strongly affected by relatively low levels of added noise and sets a conservative lower bound on the strength of the color differences likely to be detectable. For any given survey strategy and instrument, the availability of a sufficient number of measurements with sufficiently low photometric noise will determine the limiting magnitude of the asteroids to which the test can be appropriately applied.

4.4. Sensitivity to error in assumed rotational period

We evaluated the effect of assuming an incorrect rotational period in the fit photometric model. Assuming all other model parameters were accurately inferred, we adopted a period estimate drawn from a Gaussian distribution with mean equal to the true period with a variable standard deviation. Using a detection threshold of 5σ , we computed the true detection and false positive rate for the spotted and uniform synthetic asteroids respectively as a function of the standard deviation of the added uncertainty in period.

The results are shown in Figure 6. As in the previous test, the rate of false positives is not strongly dependent on error in assumed period. For all spot sizes and reflectance enhancement

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Figure 4 varies the number of observations independently in the two bands.

Selected detection rates include:

Band 1 count	Band 2 count	Detection rate
10	10	0.27

Band 1 count	Band 2 count	Detection rate
10	30	0.80
30	30	0.90
50	50	0.96
125	125	0.99

Band 2 contains the simulated spot.

More Band 2 data provide more chances to observe the feature.

However, both bands need sufficient coverage because the method compares two correlations.

The lower heatmap shows that strong imbalance can increase false positives.

For example, the most imbalanced corners reach values around 0.10 to 0.12 in this simulation.

When both filters have moderate, comparable sampling, false-positive rates are approximately zero.

The lesson is:

Filter balance and geometry coverage matter, not only total observation count.

22. Result 3 - Random measurement noise

Figure 5 plots detection rate against the standard deviation of added Gaussian magnitude noise.

As σ_m increases:

- residual scatter increases;
- true geometry-residual correlations weaken;
- true detection rate decreases;
- false-positive rate remains close to zero for independent Gaussian noise.

Noise usually destroys a real correlation rather than creating a stable surface-linked correlation.

Large spots and strong-reflectance spots remain detectable at higher noise levels.

Small spots with angular widths around 20° to 30° deteriorate rapidly.

The paper reports that detection of these small spots falls below approximately 50% for errors greater than about 0.1 mag.

Reflectance enhancements near the lower limit, around $R_f = 1.2$, are also strongly affected by modest noise.

23. Why 0.1 magnitude noise matters

For a magnitude change Δm , the flux ratio is:

$$\frac{F_2}{F_1} = 10^{-0.4\Delta m}.$$

For $\Delta m = 0.1$:

$$10^{-0.04} \approx 0.912.$$

Thus, 0.1 mag represents approximately a 9% flux difference in one direction.

That is large relative to many diluted disk-integrated surface signals.

24. Assumptions behind the probability calculation

The interpretation depends on several assumptions:

1. The two band samples are treated as independent because they are observed at different epochs.
 2. The uniform photometric model adequately removes shape, rotation, distance and phase behaviour.
 3. A real surface feature remains fixed in the asteroid's rotating frame.
 4. Residual dependence on geometry is approximately captured by a linear projection.
 5. Spearman correlation appropriately summarizes the monotonic relation.
 6. The Fisher approximation is adequate for the observation counts.
 7. Synthetic Gaussian noise approximates random measurement error.
 8. The simulated range of spot sizes and contrasts is relevant.
 9. The 5σ threshold is evaluated using the paper's stated direction and sign convention.
-

25. Important limitations

25.1 A significant result is not automatic proof

A band-dependent residual pattern may also arise from:

- filter-dependent calibration errors;
- inaccurate phase functions;
- wrong rotation period;
- wrong pole solution;
- wavelength-dependent seeing or trailing losses;
- blending with background stars;
- saturation;
- unmodelled non-convex shadowing;
- unbalanced observing geometry.

25.2 Gaussian noise is an idealization

Real survey errors can be correlated, heteroscedastic and non-Gaussian.

25.3 Multiple testing

Testing many asteroids increases the chance that at least one object passes a threshold accidentally.

Large-catalogue work should measure catalogue-level false discoveries using empirical null simulations and an appropriate multiple-testing strategy.

25.4 Direction selected from the same data

The regression uses the residual dataset to select $\hat{\mathbf{k}}$.

The same data are then used to calculate correlations.

Monte Carlo calibration helps account for this workflow, but real-data analysis should also consider split-sample or cross-apparition validation.

26. How this should be validated with real ZTF and DAMIT data

For each real asteroid:

1. Select clean ZTF g - and r -band measurements.
2. Calculate light-time-corrected geometry.
3. Generate brightness predictions using the DAMIT shape and spin model.
4. Fit appropriate band-dependent phase and magnitude parameters.
5. Calculate residuals.
6. Apply the geometry regression and Fisher correlation comparison.
7. Simulate uniform-colour data at the exact real timestamps.
8. Add noise using the actual ZTF uncertainty distribution.
9. Inject spots of different size, contrast and position.
10. Perturb the DAMIT period, pole and shape within plausible uncertainties.
11. Repeat the complete test many times.
12. Measure target-specific detection efficiency and false-positive probability.
13. Verify a candidate using separate apparitions or held-out observations.

This converts the paper's general simulation into a defensible real-object result.

27. Presentation-ready summary

The probability enters the project in five stages.

First, a uniformly coloured photometric model predicts the asteroid's brightness.

Second, observed-minus-predicted residuals reveal unexplained brightness variation.

Third, regression finds the body-fixed direction most strongly associated with those residuals.

Fourth, Spearman correlation measures the relationship between geometry and residual separately in each wavelength band.

Fifth, the Fisher transformation calculates how significantly the two band correlations differ.

The decision statistic is:

$$z_{B_1, B_2} = \frac{\operatorname{atanh}(\rho_{B_1}) - \operatorname{atanh}(\rho_{B_2})}{\sqrt{\frac{1}{n_{B_1}-3} + \frac{1}{n_{B_2}-3}}}.$$

The paper uses $z > 5$ as its candidate threshold.

Monte Carlo simulations then measure how frequently spotted asteroids are correctly detected and uniformly coloured asteroids are incorrectly flagged.

The main results are that detection improves with larger spots, stronger reflectance contrast, more observations and balanced filter coverage, while it declines with increasing photometric noise and incomplete observing geometry.

28. Formula summary

Purpose	Formula
Magnitude residual	$\Delta m_B = m_{B,\text{obs}} - m_B$
Average geometry	$\hat{\mathbf{E}} = \frac{1}{2}(\hat{\mathbf{E}}_0 + \hat{\mathbf{E}})$
Regression	$\Delta m = k_x \hat{E}_x + k_y \hat{E}_y + k_z \hat{E}_z + \epsilon$
Direction normalization	$\hat{\mathbf{k}} = \mathbf{k}/\ \mathbf{k}\ $
Geometry projection	$\mathcal{L} = \hat{\mathbf{E}} \cdot \hat{\mathbf{k}}$
Band correlation	$\rho_B = \text{corr}_{\text{Spearman}}(\mathcal{L}, \Delta m_B)$
Spearman without ties	$\rho = 1 - 6 \sum d_i^2 / [n(n^2 - 1)]$
Fisher transformation	$z_B = \frac{1}{2} \ln[(1 + \rho_B)/(1 - \rho_B)]$
Difference significance	$z_{B_1, B_2} = (z_{B_1} - z_{B_2}) / \sqrt{1/(n_{B_1} - 3) + 1/(n_{B_2} - 3)}$
Gaussian noise	$\epsilon_m \sim \mathcal{N}(0, \sigma_m^2)$
Noisy magnitude	$m_{\text{noisy}} = m_{\text{true}} + \epsilon_m$
Detection rate	$\text{TPR} = N_{\text{detected, spot}} / N_{\text{spot}}$
False-positive rate	$\text{FPR} = N_{\text{detected, uniform}} / N_{\text{uniform}}$

Reference

Humes, O., and Agarwal, J. (2026), Prospects for detecting surface color heterogeneity on asteroid surfaces from sparse multiband photometric survey data, *Astronomy and Astrophysics*, 711, A209.

- arXiv
- Published article